

lunar_lro_nac

Processed on: 2026-04-28 16:32:44

ASP version: 3.7.0-alpha

asp_plot version: 3.1.0

DEM Summary

Property	Value
DEM File	output-DEM.tif
Dimensions (px)	4720 x 4510
GSD (m)	1.04
CRS	EPSG:None
Nodata (%)	4.1
Elevation Range (m)	38.5 to 544.2

Processing Parameters

Runtime Summary

Step	Runtime
Bundle Adjust	N/A
Stereo	0 hours and 35 minutes
point2dem	0 hours and 0 minutes

Bundle Adjust Command:

Bundle adjustment not run

Stereo Command:

```
stereo M181058717LE_crop.cub M181073012LE_crop.cub M181058717LE.json M181073012LE.json --alignment-method local_epipolar  
--stereo-algorithm asp_mgm --subpixel-mode 9 out_stereo/output --corr-seed-mode 1 --sgm-collar-size 128 --compute-point-  
cloud-center-only --threads 8
```

point2dem Command:

```
point2dem --auto-proj-center --errorimage --orthoimage out_stereo/output-PC.tif out_stereo/output-L.tif
```

Report Generation Command:

```
asp_report --directory /Users/ben/Desktop/asp-plot-examples/lunar_lro_nac/ --stereo-directory out_stereo/ --no-basemap  
--altimetry-csv /Users/ben/Desktop/asp-plot-examples/lunar_lro_nac/lola/LolaRDR-10N-10N_15E15E_20260428T183313426_pts_csv.csv --no-geometry --subset-km 0.5  
--reuse-selections /Users/ben/Desktop/uw-github/asp_plot/reports/LRO_NAC-asp-report_figure_selections.yml --report-  
filename /Users/ben/Desktop/uw-github/asp_plot/reports/LRO_NAC-asp-report.pdf
```

Input Scenes

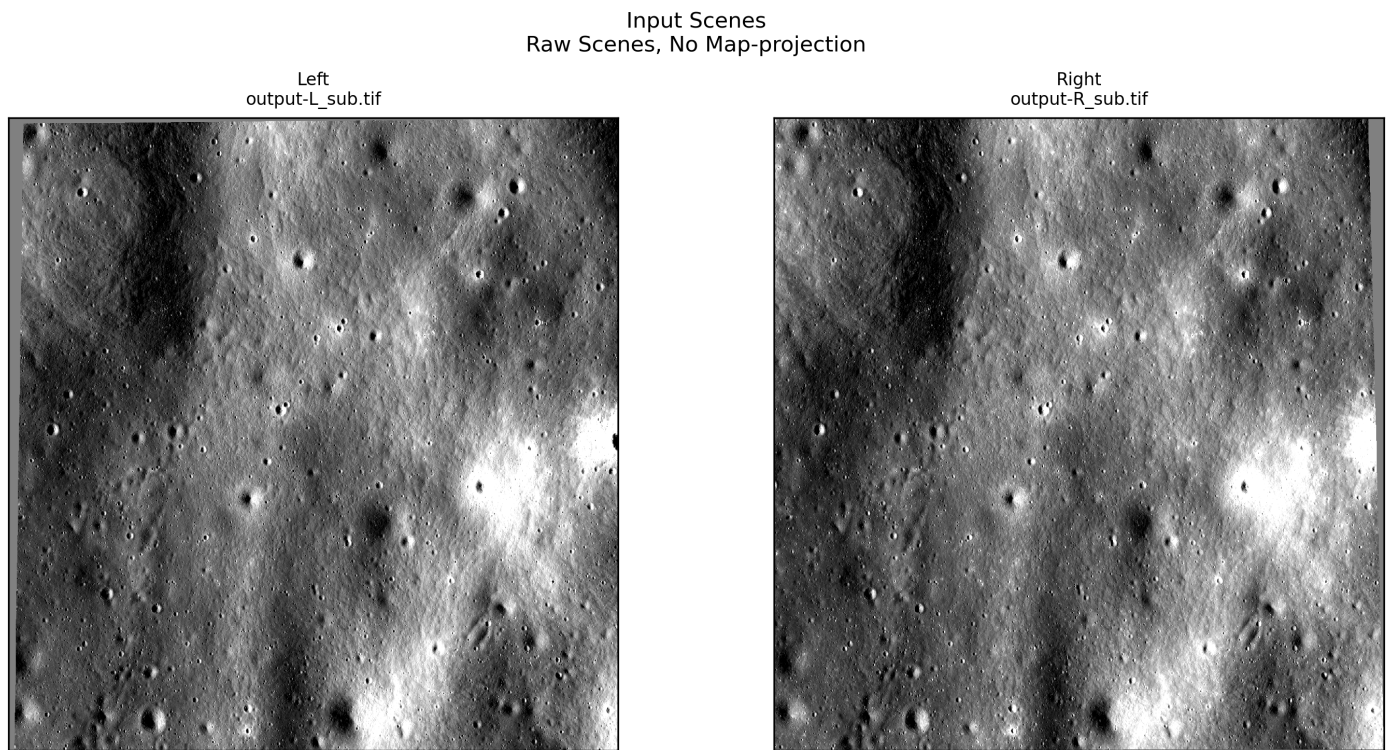


Figure 1: Left and right input scenes used for stereo processing. Non-mapprojected scenes are shown after ASP's alignment step (e.g., affineepipolar), which rotates images to create horizontal epipolar lines for correlation. Mapprojected scenes have been orthorectified with RPCs against a reference DEM to roughly align the two images prior to correlation, which reduces the disparity search range; they are displayed here in their map-projected orientation.

Match Points

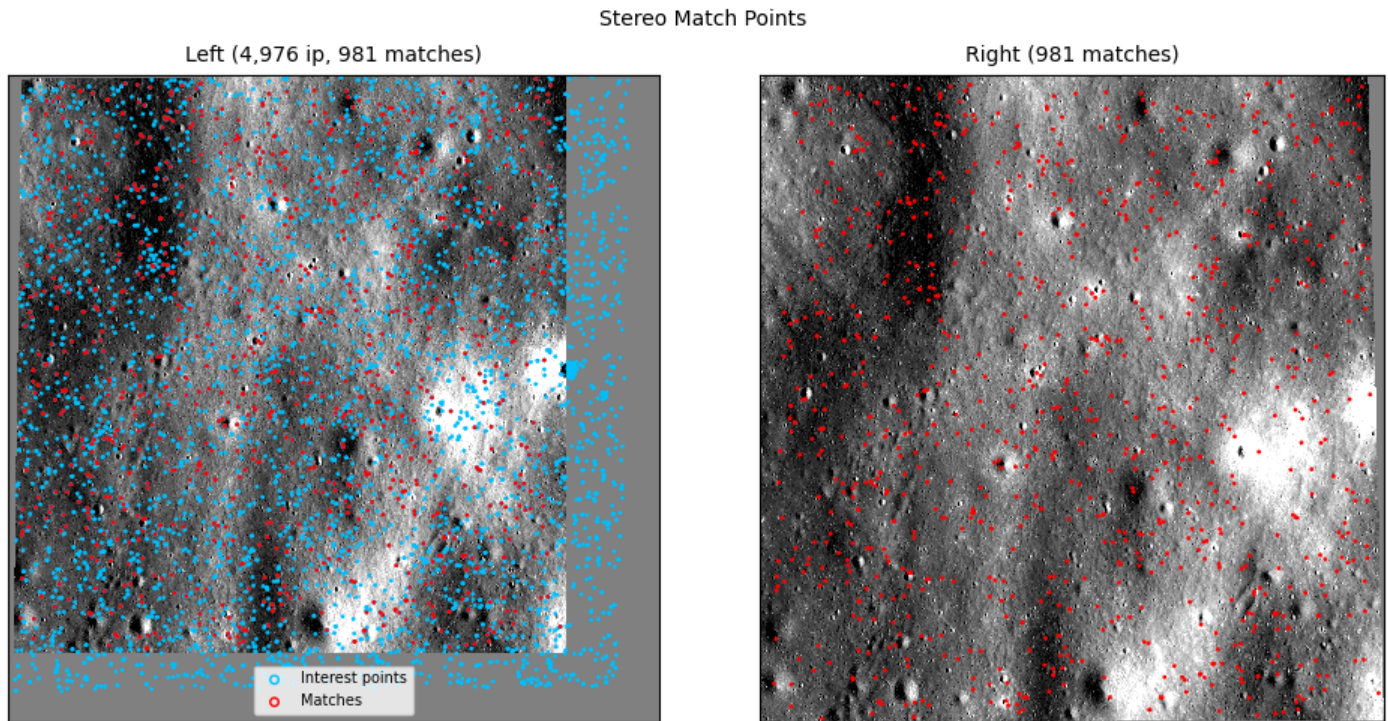


Figure 2: Interest point matches between left and right images (red), overlaid on the raw interest points detected on each image individually (blue, when the .vwip files are present in the stereo directory). These are produced by stereo_corr during its initial interest point matching step, which is used to set the search windows for subsequent dense correlation (not the dense correlation matches themselves). Comparing the raw interest points against the matches shows whether sparse matching traces back to poor matching or to areas with no detected interest points at all (e.g., limited texture, clouds, water). Some ASP configurations only write a .vwip file for the left image (the right image's interest points are detected on a temporary aligned image and not saved), in which case only matches are shown on the right.

Disparity

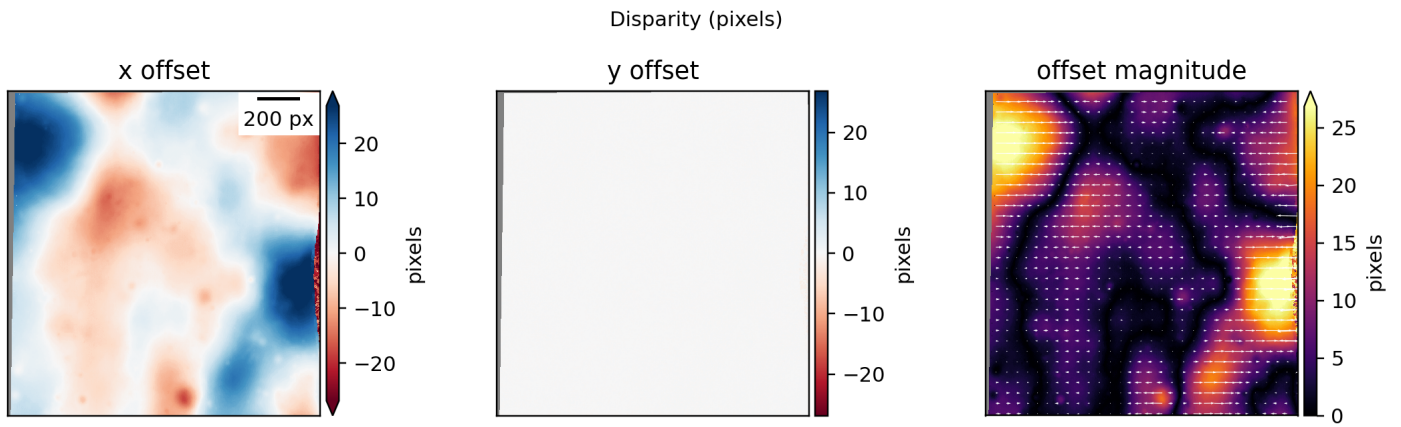


Figure 3: Horizontal and vertical disparity maps in pixels with quiver overlay.

DEM Results

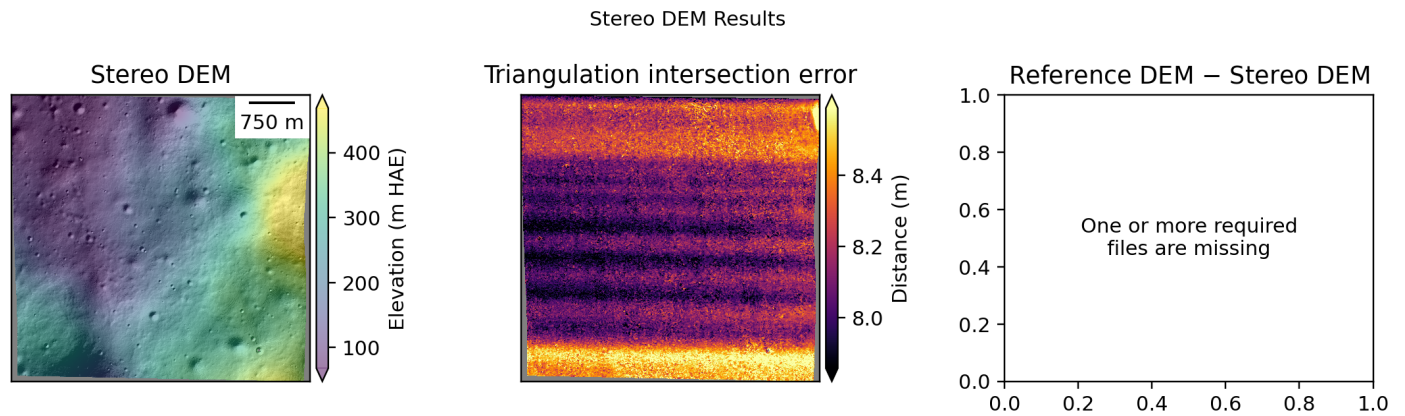


Figure 4: Output DEM with intersection error map and difference relative to the reference DEM used in processing.

Detailed Hillshade

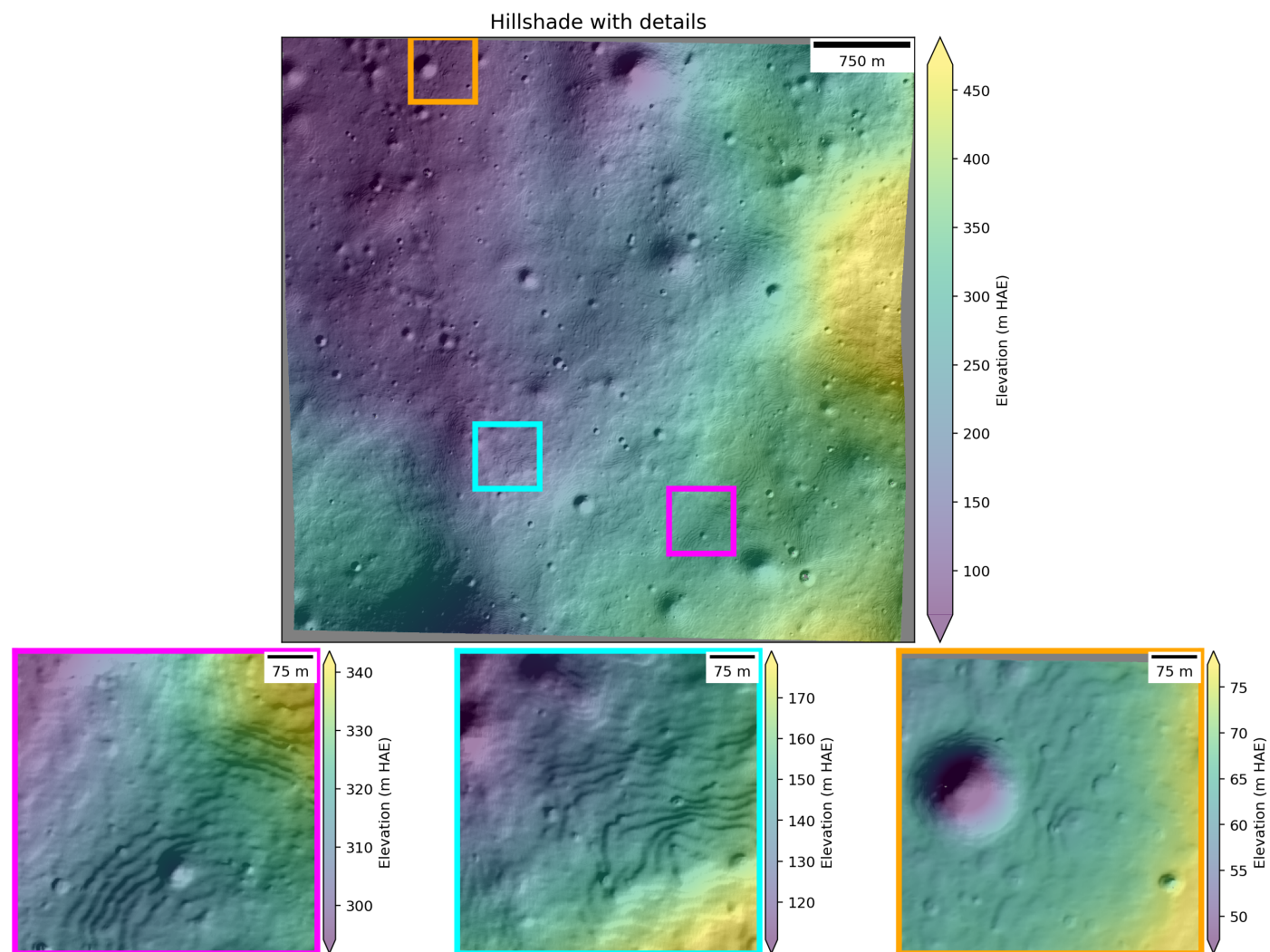


Figure 5: DEM hillshade. If the intersection error is available, zoomed subsets selected from low, medium, and high (left to right) uncertainty areas are displayed in the second row. If the mapprojected image is available, corresponding ortho image subsets are displayed in the bottom row.

LOLA Altimetry Map

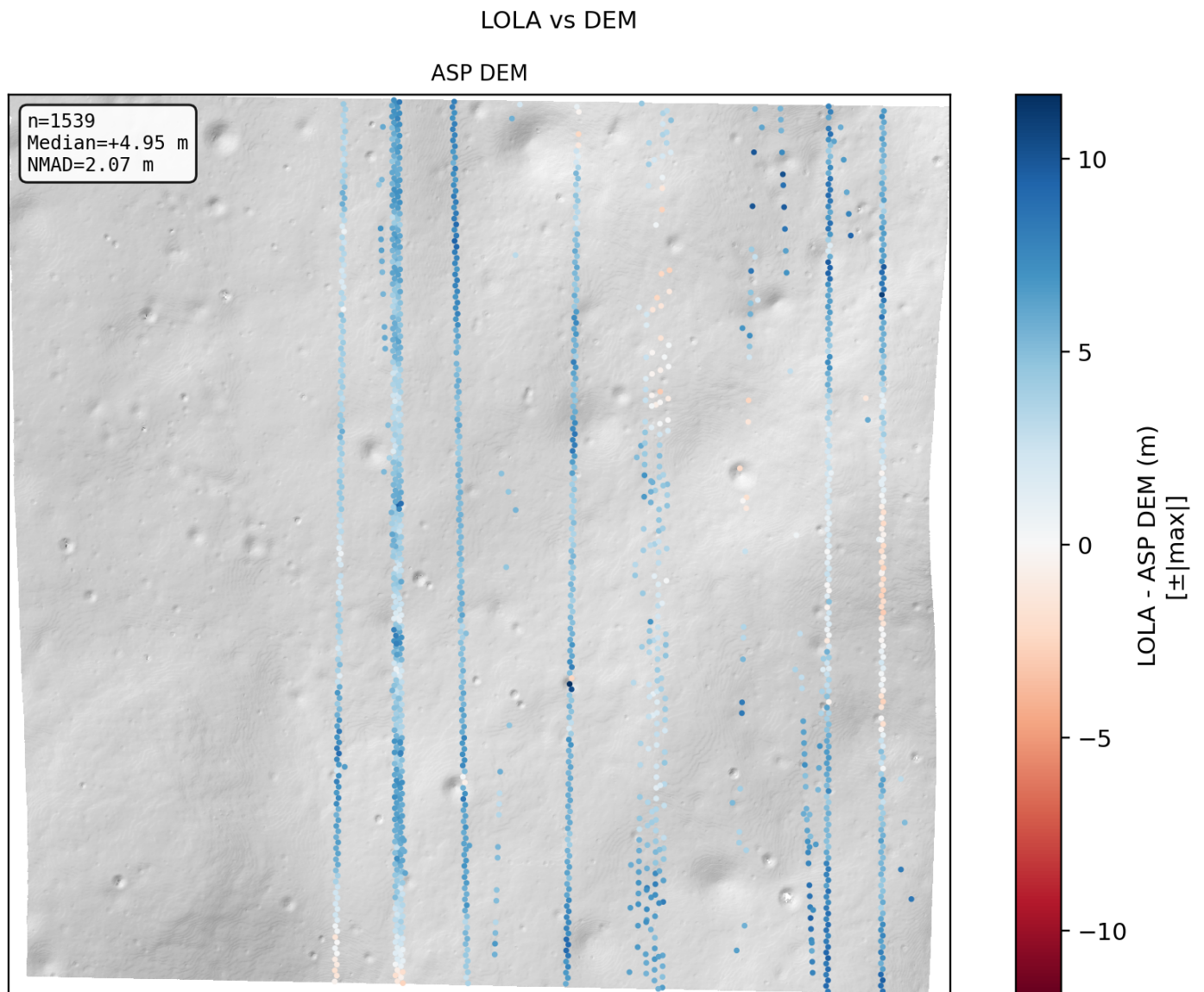


Figure 6: LOLA elevation differences vs. ASP DEM.

LOLA Altimetry Histogram

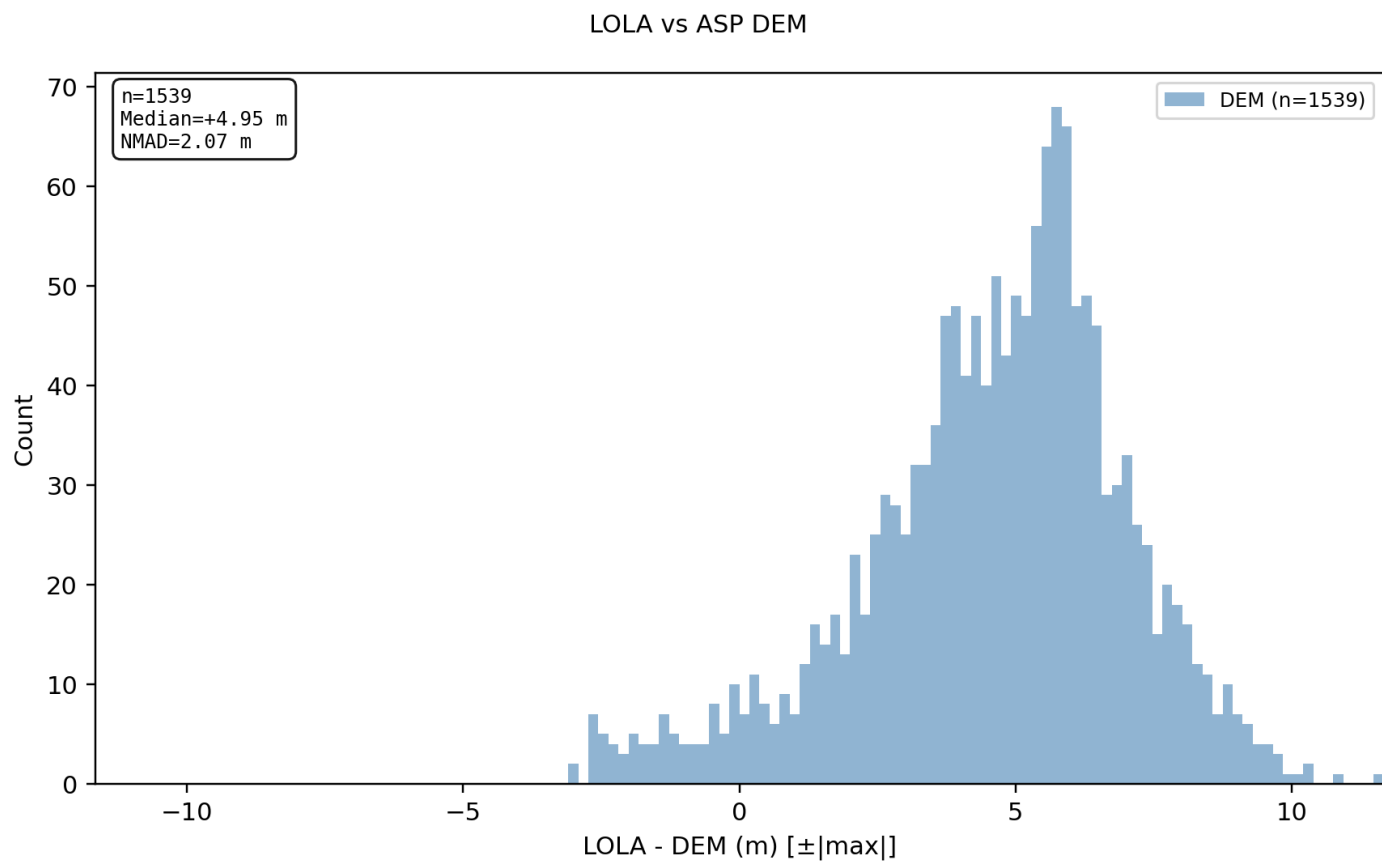


Figure 7: Distribution of elevation differences between LOLA and ASP DEM.

DEM Alignment with LOLA

Alignment Parameters

Parameter	Value
max_displacement	500
minimum_points	20
min_translation_threshold	0.1
improvement_threshold_pct	5.0

Error Statistics (m)

Statistic	Before	After	Change
Median	5.71	2.01	-64.7%
NMAD	3.21	2.07	-35.5%
RMSE	153	153	+0.2%

Translation (m)

Component	Value
North	-0.09
East	-1.18
Down	4.76
Magnitude T	4.90

ASP's `pc_align` estimates a rigid 3D translation that minimizes the height residuals between the ASP DEM and the LOLA planetary radii. The CSV is passed as the reference cloud with `--csv-format '1:lon 2:lat 3:radius_m'`, and `--datum` is set to `D_MARS` or `D_MOON` to match the ASP DEM. The resulting translation is applied to the DEM directly (geotransform + pixel-value shift, no resampling) to produce the aligned DEM.

Alignment Parameters (above):

- `max_displacement`: `pc_align` upper bound on the translation magnitude (m).
- `minimum_points`: minimum LOLA points that overlap the DEM; below this the alignment is skipped.
- `min_translation_threshold`: minimum translation magnitude (as a fraction of the DEM GSD) required to write out an aligned DEM.
- `improvement_threshold_pct`: minimum percentage reduction in the median residual required to keep the aligned DEM on disk; below this, the aligned DEM is removed.

Error Statistics (above, in meters): each row is one statistic of the DEM-vs-LOLA absolute height residuals, Before and After alignment; Change is the percent change from Before to After, so negative is an improvement.

- Median: median residual.
- NMAD: normalized median absolute deviation; a robust spread, insensitive to outliers.
- RMSE: root-mean-square residual; outlier-sensitive, so a large RMSE next to a small NMAD flags a tail of gross errors rather than a broad misfit.
- `p16` / `p50` / `p84`: shown instead of the above when the `pc_align` log predates ASP 3.7.0 (which added the Median / NMAD / RMSE summary): 16th / 50th / 84th percentile of the residuals; `p50` is the median.

Translation (above, in meters): the applied translation vector as North / East / Down components and its magnitude |T|. Positive Down means the DEM was too high.

Median improved from 5.71 m -> 2.01 m (64.7% reduction). Aligned DEM written to /Users/ben/Desktop/asp-plot-examples/lunar_lro_nac/out_stereo/output-DEM_pc_align_translated.tif.

LOLA Altimetry Map (Aligned DEM)

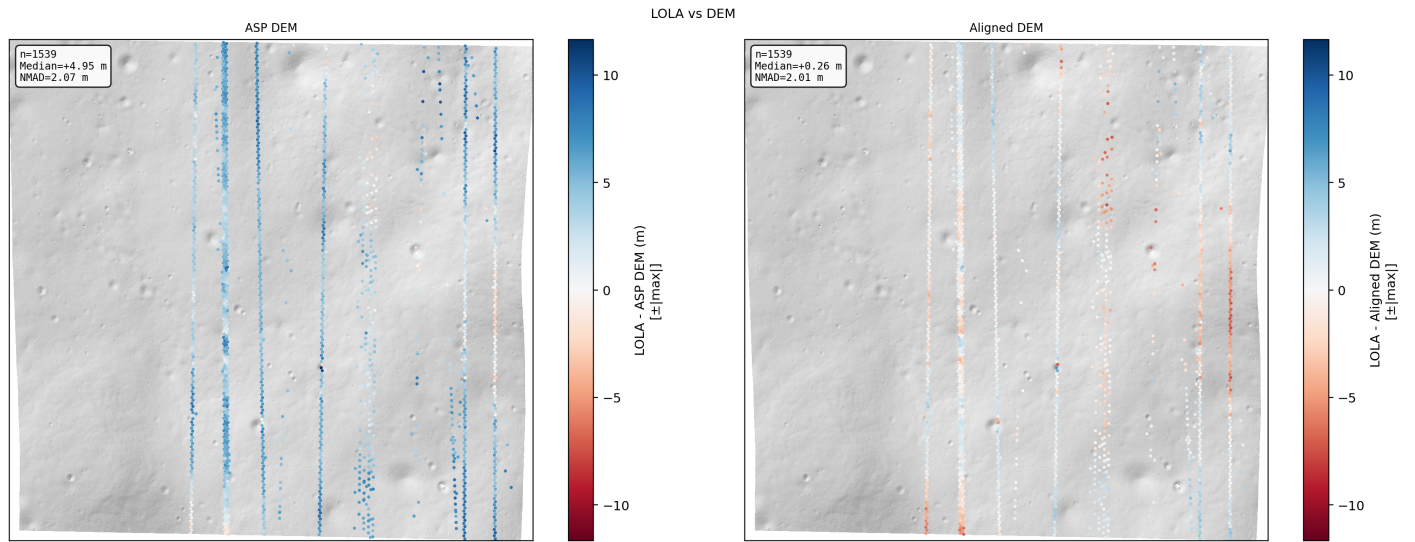


Figure 9: Pre- (left) and post-alignment (right) map views of LOLA elevation differences. The aligned-DEM hillshade is used as the backdrop for both panels.

LOLA Altimetry Histogram (Aligned DEM)

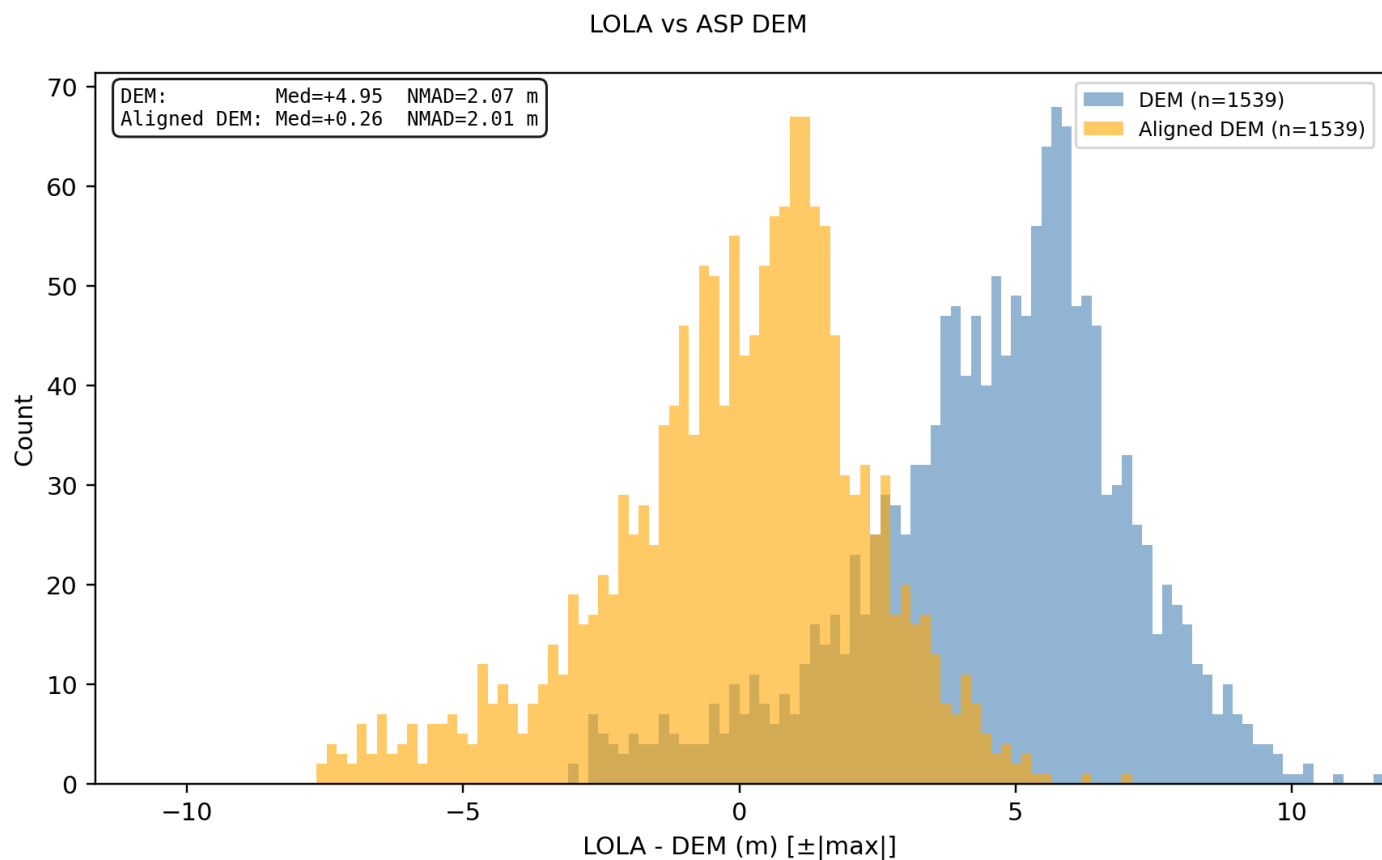


Figure 10: Pre- (steelblue) and post-alignment (orange) distributions of LOLA minus DEM height differences with shared bin edges.